The Relationship between Visual Disability and Visual Field in Patients with Glaucoma

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Purpose: To evaluate the relationship between visual disability and the remaining visual field in glaucoma patients.

Design: Prospective noncomparative survey.

Participants: One hundred forty-seven Japanese patients with glaucoma were examined.

Methods: Using a previously developed questionnaire, we assessed the relationship between visual disability indices and both the visual field examined using program 30-2 of the Humphrey Field Analyzer and visual acuity.

Main Outcome Measures: Pearson's correlation coefficients between visual disability indices and the 22 visual field and 2 visual acuity measurements were calculated. Multiple regression analysis with stepwise variable selection identified factors significantly contributing to visual disability.

Results: We identified a significant correlation between visual disability and both the visual field and visual acuity measurements collected. Of the factors examined, retinal sensitivity in the lower hemifield within 5° of the fixation and visual acuity in the better eye significantly contributed to all the item and total disability indices (DIs), measurements of the degree of visual disability as evaluated by the questionnaire; visual acuity in the worse eye contributed to two of the seven item DIs, as well as the total DI. The R value for total DI ($R = 0.87$, $P < 0.0001$) indicated that approximately 76% of the visual disability could be explained by these factors.

Conclusions: The Japanese glaucoma patients' visual disability is explained primarily by retinal sensitivity in the lower hemifield within 5° of the fixation and visual acuity in the better eye and secondarily by visual acuity in the worse eye. *Ophthalmology 2003;110:332–339 © 2003 by the American Academy of Ophthalmology.*

It is estimated that 150 million people are visually disabled in the world today,[1] including approximately 307,000 people in Japan.[2] According to the World Health Organization,[3] the main causes of visual impairment are cataract, trachoma, glaucoma, and chorioretinal diseases. In developing nations, cataract and trachoma, both treatable diseases, are the main causes of visual impairment. In developed countries, however, glaucoma and other chorioretinal or optic nerve diseases are the primary afflictions. Unfortunately, there are no treatments available which completely avert the onset of visual impairment caused by these diseases despite recent medical advances.

Those with low vision confront visual disability in their daily and social lives, leading to unavoidable deterioration in overall quality of life. With appropriate rehabilitation allowing these patients to use the remaining visual abilities, this disability can be lessened to permit independent lifestyles.[4]

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In providing rehabilitation, we first need to understand the patient's perception of visual disability in daily life. Because visual disability results in changes in individual skills and abilities, the use of questionnaires is the most accurate and useful means by which to assess visual disability. Although multiple questionnaires[5–12] have been developed in Western countries, most of these questionnaires[5–11] aim to assess visual disability in patients with visual acuity impairment. Visual acuity alone may not be an accurate predictor of visual function status.[8,13] Few questionnaires have been developed to assess visual disability in patients with visual field impairment but with relatively intact visual acuity; Mills and Drance[12] created a questionnaire intended to assess visual disability in patients with severe visual field impairment. Glaucoma is one of the main causes of visual impairment, both in Japan and throughout the world.[3] Often in glaucoma, central visual acuity is retained until the late stages of the disease, despite severe constriction of the visual field.[14] Because many of the problems resulting from visual field impairment may differ markedly from those caused by visual acuity impairment, development of a reliable questionnaire aimed at this purpose is indispensable for the accurate assessment of visual disability caused by visual field impairment. Moreover, because most publications in Japan and other Asian countries, including newspapers, books, and magazines, are printed with the text reading vertically, questionnaires developed in Western countries may not be appropriate for

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Japanese patients. No questionnaires assessing visual disability in patients who read and write vertically have been published.

Although visual disability varies widely depending on the location of the visual field impairment despite similar remaining visual field areas,[15] no report has compared the relationship between visual field and visual disability on the basis of the location of the visual field impairment in patients with visual field impairment.

We previously developed a preliminary questionnaire[16] based on interviews with 120 glaucoma patients to assess visual disability in Japanese patients who read and write vertically.

After the evaluation of patients with visual field damage who retain decent central visual acuity, we found that this preliminary questionnaire was useful in evaluating the visual disability of patients with retinitis pigmentosa, a representative disease causing visual field impairment.[17] We were then able to exclude factors unrelated to visual impairment by analysis of principal components and determination of factors contributing to disease. The resulting revised questionnaire was found both to be internally consistent and reliable.

Using the revised questionnaire, this survey sought to evaluate the regions of the central visual field most important for quality of vision in patients with glaucoma.

Subjects and Methods

Subjects

We evaluated visual disability in 147 patients (75 male and 72 female) with glaucoma (69 patients with primary open-angle glaucoma [POAG], 56 patients with normal tension glaucoma [NTG], and 22 patients with primary angle-closure glaucoma). All patients involved in the study fulfilled the following criteria: (1) glaucoma was the only disease leading to visual field damage and/or visual acuity impairment; (2) patients were followed for more than 6 months at either Tokyo Teishin Hospital or University of Tokyo Hospital; (3) visual field was evaluated using program 30-2 of the Humphrey Field Analyzer (HFA 30-2; Carl Zeiss, Dublin, CA) with reliable (fixation loss $\leq$ 33%, false-negative or false-positive $\leq$ 33%) results; (4) patients lacked other physical impairments; and (5) visual field damage was clinically stable with medically or surgically controlled intraocular pressure for at least 2 years. Clinical diagnosis of NTG was made in open-angle glaucoma patients with a recorded intraocular pressure consistently at approximately 21 mmHg with 24-hour fluctuations.

The mean ($\pm$standard deviation) age of patients was 61.5 $\pm$ 11.9 years, ranging from 25 to 85 years. The mean of the best-corrected visual acuity in the logarithm of the minimum angle of resolution was 0.3 $\pm$ 0.5 (range, -0.2 to 2.0) in the eye with better visual acuity (better eye), 0.9 $\pm$ 0.8 (range, -0.2 to 2.0) in the eye with worse visual acuity (worse eye). The mean deviation (MD) of HFA 30-2 was -11.2 ± 5.9 (range, -31.0 to $+2.9$) dB in the better eye (better MD) and -17.4 ± 6.5 (range, -32.2 to -0.1) dB in the worse eye (worse MD). Sixty-four patients worked regularly, whereas 83 patients were currently out of work. No significant difference was seen between either the male and female groups or the working and nonworking groups in HFA 30-2 measurements.

Materials and Methods

Relationship between Visual Disability Assessed by the Revised Questionnaire and Visual Acuity

Visual disability in the 147 glaucoma patients was assessed using the revised questionnaire. The questionnaire, originally written in Japanese, contained 30 questions addressing 7 items: legibility of letters (letters), sentences, walking, using public transportation (going out), dining, dressing, and additional miscellaneous activities (miscellaneous) (Table 1, excluding the questions, written with italics). Each question had three possible responses, scored as follows, greatly disabled, 2 points; slightly disabled, 1 point; and not disabled, 0 point. We calculated the visual disability index (DI) for each item (item DI: the mean score given by the questions within each item) and their sum (total DI) for each patient.

We first assessed the relationship between patients' visual disability, quantified by the revised questionnaire, and visual acuity. Because visual acuity differed from eye to eye, we compared the relationship between patients' visual disability and two representative visual acuity measurements: visual acuity in the logarithm of the minimum angle of resolution in the better eye and in the worse eye. We then calculated Pearson's correlation coefficients between the above individual visual acuity measurements and both the item and total DIs. Multiple regression analysis was also applied using total DI as a dependent variable and the two representative visual acuity measurements as independent variables.

Relationship between Visual Disability Assessed by the Revised Questionnaire and Remaining Visual Field

Within 1 month of visual disability assessment, we tested the monocular central 30° visual field using HFA 30-2 in both eyes with a 5-minute rest between each eye examination. Next, we studied the relationship between the patient's visual disability, quantified by the revised questionnaire, and the remaining visual field. To determine the regions within the central 30° visual field correlated with visual disability indices, the central 30° visual field was divided into the following subfields: the visual field within 10° of the fixation (VF_{10}), the visual field between 11° and 20° of the fixation (VF_{11-20}), the visual field between 21° and 30° of the fixation (VF_{21-30}), and the entire central 30° visual field (VF_{30}). Test points within the central 10° of the fixation were further divided into six clusters (Fig 1 A–F; $VF_A - VF_F$). The average sensitivities (dB) in the preceding subfields or clustered test points in the binocular visual field were calculated using two independent methods. First, the most sensitive point determined from the right and left corresponding visual field test point was selected and designated as the sensitivity of each corresponding visual field test point (HFA 30-2). These values were further averaged over the subfield or test point cluster (the highest sensitivity). Second, the least sensitive point from the right and left corresponding visual field test point was selected and adopted as the sensitivity of each corresponding visual field test point. These values were then averaged over the subfield or test point cluster (the lowest sensitivity). The better MD and worse MD of HFA 30-2 were used as visual field measurements representing overall visual field loss. We also calculated Pearson's correlation coefficients for the above 22 individual visual field measurements and item and total DIs. Multiple regression analysis was also applied using total DI as a dependent variable and the representative visual field measurements as independent variables.

Table 1. Questions Included in the Preliminary Questionnaire and the Revised Questionnaire (Excluding the Questions Written in Italics)

Legibility of letters: letters	
1. Can you read the headlines of a newspaper?	(Yes/With difficulty/No)
2. Can you read small print in a newspaper?	(Yes/With difficulty/No)
3. Can you read words in a dictionary?	(Yes/With difficulty/No)
4. Can you see the numbers in a telephone directory?	(Yes/With difficulty/No)
5. Can you make out a fare table for trains and subways?	(Yes/With difficulty/No)
Sentences	
6. Do you have difficulty reading and writing?	(No/Occasionally/Frequently)
7. When you write sentences in vertical lines, does it lean to either direction?	(No/Occasionally/Frequently)
● *Can you read movie subtitles?*	*(Yes/With difficulty/No)*
8. When you read, can you find the next line easily?	(Yes/With difficulty/No)
Walking	
9. Do you have difficulty walking because of your visual problems?	(No/Occasionally/Frequently)
10. Can you take a walk by yourself?	(Yes/With difficulty/No)
11. Do you misjudge traffic signals?	(No/Occassionally/Frequently)
12. Do you bump into people or objects while walking?	(No/Occasionally/Frequently)
13. Do you stumble on the stairs?	(No/Occasionally/Frequently)
14. Do you fail to notice changes in the ground?	(No/Occasionally/Frequently)
15. Do you fail to recognize your friends until they talk to you?	(No/Occasionally/Frequently)
16. Do you fail to see people or cars approaching you from the side?	(No/Occasionally/Frequently)
Going out	
17. Do you have difficulty going out because of your visual problems?	(No/Occasionally/Frequently)
18. Do you need somebody to accompany you to go to new places?	(No/Preferably/Yes)
19. Can you get a cab by yourself?	(Yes/With difficulty/No)
20. Do you have difficulty traveling by train?	(No/Occasionally/Frequently)
21. Do you feel uneasy going out at night because of your visual problems?	(No/Occasionally/Frequently)
Dining	
22. Do you have difficulty dining because of your visual problems?	(No/Occasionally/Frequently)
● *Can you dine by yourself?*	*(Yes/With difficulty/No)*
23. Do you drop food while dining because of your visual problems?	(No/Occasionally/Frequently)
24. Do you spill tea while pouring into a cup?	(No/Occasionally/Frequently)
25. Do you have difficulty using chopsticks?	(No/Occasionally/Frequently)
Dressing	
● *Can you change clothes by yourself?*	*(Yes/With difficulty/No)*
● *Do you sometimes put underwear on inside out?*	*(No/Occasionally/Frequently)*
26. Do you ever button up clothing in the wrong order?	(No/Occasionally/Frequently)
● *Do you have difficulty dressing because of your visual problems?*	*(No/Occasionally/Frequently)*
27. Can you see your face clearly in the mirror?	(Yes/With difficulty/No)
Miscellaneous	
28. Can you recognize people's faces on TV?	(Yes/With difficulty/No)
29. Do you have difficulty finding objects dropped on the floor?	(No/Occasionally/Frequently)
30. Do you have difficulty dialing the telephone?	(No/Occasionally/Frequently)

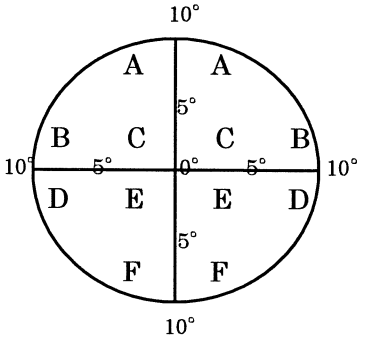


Figure 1. Clustered test points (A–F) within the central 10° of the fixation. Test points in the central 10° of the fixation were divided into six clusters (A, B, C, D, E, and F).

Relationship between Visual Disability Assessed by the Revised Questionnaire and Visual Acuity and Visual Field

Multiple regression analysis was applied to the measurements obtained in the previous two studies using the following dependent and independent variables. Seven item DIs and total DI were used as dependent variables for comparison with the independent variables of patient gender, age, and occupation, 2 visual acuity measurements (better eye and worse eye) and 22 visual field measurements (VF_{10}, $VF_{11–20}$, $VF_{21–30}$, VF_{30}, VF_A, VF_B, VF_C, VF_D, VF_E, and VF_F for both the highest and lowest sensitivities, and better and worse MD). Independent variables were selected for each dependent variable for analysis by the stepwise variable selection ($P < 0.05$).

All the statistical calculations were carried out using Statistical Analysis Software, System version 6 (SAS Institute, Cary, NC).

Table 2. Pearson's Correlation Coefficients between the Individual Visual Acuity Measurements and Visual Disability Indices

	Item Disability Index							Total Disability Index
	Letters	*Sentences*	*Walking*	*Going Out*	*Dining*	*Dressing*	*Miscellaneous*	
Better*	0.73†	0.68†	0.67†	0.70†	0.64†	0.63†	0.71†	0.74†
Worse‡	0.51	0.50	0.50	0.51	0.49	0.46	0.56†	0.56†

*Visual acuity in logarithm of the minimum angle of resolution in an eye with better visual acuity than the fellow eye.
†$P < 0.0001$.
‡Visual acuity in logarithm of the minimum angle of resolution in an eye with worse visual acuity than the fellow eye.
For all correlation coefficients, correlation was significant at $P < 0.0005$.

Results

Relationship between Visual Disability Assessed by the Revised Questionnaire and Visual Acuity

Visual acuity in the better and worse eye correlated significantly with all the disability indices ($r = 0.46$ to 0.74) (Table 2). Visual acuity in the better eye demonstrated the highest correlation with the item and total DIs ($r = 0.63$ to 0.74).

Using stepwise variable selection, the visual acuities in both the better and worse eyes were found to be significant predictors of total DI ($P < 0.0001$, $P < 0.001$). The standardized partial regression coefficients for visual acuity in the better and worse eyes were 0.65 and 0.21, respectively. The *R* value determined between the predicted and actual scores of total DI was 0.78 ($P < 0.0001$).

Relationship between Visual Disability Assessed by the Revised Questionnaire and Remaining Visual Field

Regardless of the selection of either highest or lowest sensitivity in the calculation of VF_{10}, $VF_{11–20}$, $VF_{21–30}$, VF_{30}, VF_A, VF_B, VF_C, VF_D, VF_E, and VF_F, all these visual field measurements exhibited significant correlations with the DIs ($r = -0.70$ to -0.30) (Table 3). The better and worse MDs also demonstrated significant correlations with the DIs ($r = -0.64$ to -0.44). Of these, the highest sensitivity VF_E, corresponding to the sensitivity of the lower hemifield within 5° of the fixation, possessed the best correlation with the item and total DIs ($r = -0.70$ to -0.55) (Fig 2).

By stepwise variable selection, however, only the highest sensitivity VF_E was a statistically significant independent predictor for total DI (the standardized partial regression coefficients: −0.57, $P < 0.0001$). None of the other independent variables were selected as predictors for total DI. The *R* value determined between the predicted scores and the actual scores of total DI was 0.73 ($P < 0.0001$).

Relationship between Visual Disability Assessed by the Revised Questionnaire and Visual Acuity and Visual Field

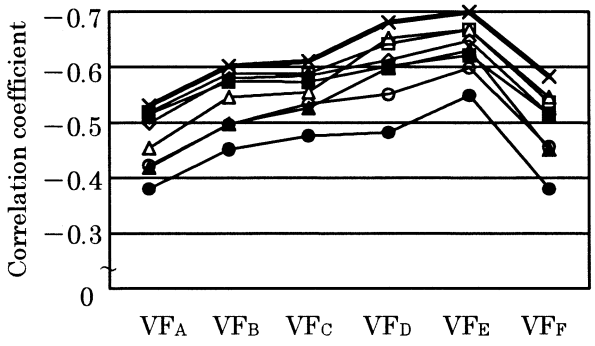
By stepwise variable selection, only the highest sensitivity VF_E ($P = 0.0007$ to 0.0417) and visual acuity in the better eye ($P < 0.0001$) possessed significant correlations with all the dependent variables (Table 4). Visual acuity in the worse eye ($P = 0.0046$ to 0.0289) also possessed significant correlations with dining, miscellaneous, and total DI. None of the other independent variables

Table 3. Pearson's Correlation Coefficients between the Individual Visual Field Measurements and Visual Disability Indices

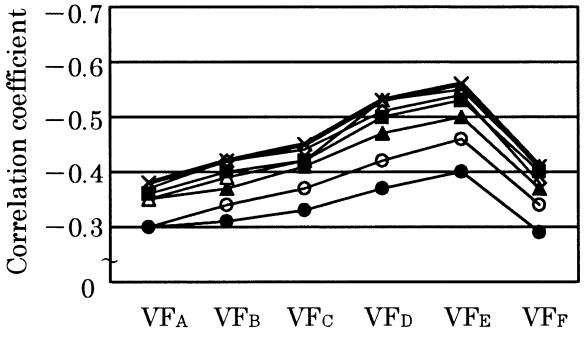
	Item Disability Index							Total Disability Index
	Letters	*Sentences*	*Walking*	*Going Out*	*Dining*	*Dressing*	*Miscellaneous*	
VF_{30} (visual field within 30° of the fixation)								
Highest[1]	−0.56*	−0.59*	−0.61*	−0.64*	−0.54*	−0.48*	−0.58*	−0.63*
Lowest[2]	−0.51*	−0.55*	−0.59*	−0.59*	−0.51*	−0.43*	−0.54*	−0.60*
VF_{10} (visual field within 10° of the fixation)								
Highest[1]	−0.60*	−0.65*	−0.62*	−0.66*	−0.56*	−0.50*	−0.64*	−0.67*
Lowest[2]	−0.39	−0.42	−0.42	−0.40	−0.34	−0.30	−0.41	−0.43*
$VF_{11–20}$ (visual field between 11° and 20° of the fixation)								
Highest[1]	−0.52*	−0.56*	−0.61*	−0.62*	−0.51*	−0.46*	−0.56*	−0.60*
Lowest[2]	−0.33	−0.35	−0.35	−0.33	−0.30	−0.30	−0.34	−0.36
$VF_{21–30}$ (visual field between 21° and 30° of the fixation)								
Highest[1]	−0.51*	−0.53*	−0.59*	−0.61*	−0.48*	−0.44*	−0.52*	−0.58*
Lowest[2]	−0.31	−0.33	−0.34	−0.32	−0.30	−0.30	−0.31	−0.34
Mean deviation								
Better[3]	−0.57*	−0.60*	−0.61*	−0.63*	−0.55*	−0.49*	−0.60*	−0.64*
Worse[4]	−0.50*	−0.55*	−0.58*	−0.60*	−0.50*	−0.44*	−0.52*	−0.59*

[1]The most sensitive point and [2]the least sensitive point of the right and left corresponding visual field test point was selected when calculating each visual field measurement.
[3]Mean deviation (MD) in the eye with better MD.
[4]Mean deviation (MD) in the eye with worse MD.
For all correlation coefficients, correlation was significant at $P < 0.005$.
*$P < 0.0005$.

1.



2.



3.

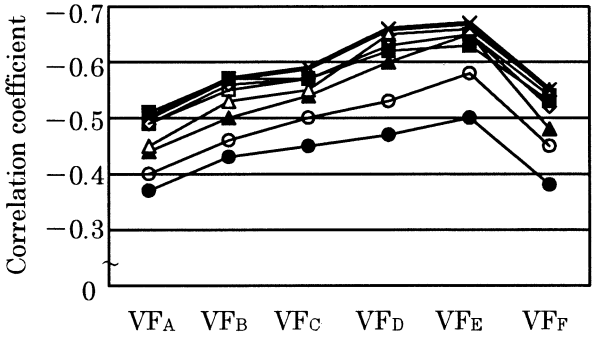


Figure 2. Pearson's correlation coefficients between VF_A − VF_F (the average sensitivities over the clustered test points A–F, which correspond to Fig 1) and visual disability indices. **1,** The highest sensitivity; **2,** the lowest sensitivity**; 3,** the mean sensitivity. Item disability index (DI): Letters, ▲; sentences, △; walking, ■; going out, □; dining, ○; dressing, ●; miscellaneous, ◇; total DI, ×.

were significantly correlated with any of the disability indices. The standardized partial regression coefficients for the highest sensitivity VF_E, the visual acuity in the better eye, and the visual acuity in the worse eye were −0.27 to −0.21, 0.34 to 0.42, and 0.16 to 0.21, respectively. The *R* values determined between the predicted scores and the actual scores of item and total DIs were 0.72 to 0.84 and 0.87, respectively ($P < 0.0001$).

Discussion

Previous questionnaires[5–12] assessing visual disability in patients with visual acuity or field impairment unanimously covered the seven items examined in our questionnaire: letters, sentences, walking, going out, dining, dressing, and

Table 4. The Results of the Multiple Regression Analysis—Independent Variables: Visual Acuity and Visual Field Measurements

Dependent Variables	Standardized Partial Regression Coefficient (*P* Value)			*R* Value (*P* Value)
	Visual Acuity (Logarithm of the Minimum Angle of Resolution)			
	Better Eye	*Worse Eye*	VF_E* *Highest Sensitivity*	
Item Disability Index				
Letters	0.51 (<0.0001)	NS	0.24 (0.0007)	0.83 (<0.0001)
Sentences	0.34 (<0.0001)	NS	−0.23 (0.0392)	0.83 (<0.0001)
Walking	0.38 (<0.0001)	NS	−0.22 (0.0417)	0.78 (<0.0001)
Going out	0.41 (<0.0001)	NS	−0.25 (0.0035)	0.84 (<0.0001)
Dining	0.39 (<0.0001)	0.21 (0.0046)	−0.27 (0.0002)	0.72 (<0.0001)
Dressing	0.36 (<0.0001)	NS	−0.22 (0.0055)	0.72 (<0.0001)
Miscellaneous	0.41 (<0.0001)	0.17 (0.0289)	−0.21 (0.0013)	0.83 (<0.0001)
Total disability index	0.42 (<0.0001)	0.16 (0.0067)	−0.24 (0.0057)	0.87 (<0.0001)

*The averages of the sensitivities over the clustered test points E. (Clustered test points E corresponds to Fig 1).
NS = not significant; VF = visual field.

miscellaneous. Driving, however, an important section in questionnaires developed in Western countries, was not included in this questionnaire. None of the patients involved in this study complained of any inconvenience in driving. In Japan, a driving license is not issued to a person with a best-corrected visual acuity less than 0.7, meaning that many of the examined subjects cannot drive by themselves. Furthermore, in big Japanese cities such as Tokyo, many do not or need not drive by themselves.

Because no published reliable questionnaire has been developed to assess visual disability in patients with visual field damage who read and write vertically, our revised questionnaire may also be of clinical use in Taiwan and China, where people read and write vertically.

By use of the revised questionnaire, we evaluated the relationship between visual disability and the visual field assessed using program HFA 30-2 and visual acuity in 147 patients at various stages of advanced glaucoma.

Among the 22 visual field measurements (VF_{10}, $VF_{11–20}$, $VF_{21–30}$, VF_{30}, VF_A, VF_B, VF_C, VF_D, VF_E, and VF_F of the highest and lowest sensitivities and better and worse MDs) and two visual acuity measurements (visual acuity in the better eye and visual acuity in the worse eye) made, multiple regression analysis with stepwise variable selection determined that only the highest sensitivity VF_E and visual acuity in the better eye demonstrated significant contribution to all the item and total DIs measurements of the degree of visual disability as evaluated by our questionnaire. Visual acuity in the worse eye exhibited a significant contribution to two of the seven item DIs, as well as the total DI. Crabb et al[18] reported that use of the highest sensitivity from either eye for each corresponding test location simulated the binocular visual field sensitivity of glaucoma patients well. Nelson-Quigg et al[19] also reported that either the Best Location model (in which binocular visual field sensitivity is predicted to be the most sensitive of two visual field locations between eyes for corresponding visual field points) or the Binocular Summation model (in which binocular visual field sensitivity is predicted by probability summation of the two eyes' sensitivities) provided the best estimates for binocular visual field sensitivity. Thus, although binocular visual field was not examined in this study, the present visual field measurements of the highest sensitivity estimates with reasonable accuracy the binocular visual field of patients most essential in patients' daily lives. The high *R* values for the regression, 0.87 for total DI and between 0.72 and 0.84 for item DIs, indicate that visual disability can be explained by the combination of the previous three parameters, primarily the first two. Thus, maintenance of both the visual acuity and the visual field in the lower hemifield within 5° of the fixation are important for quality of vision in Japanese glaucoma patients.

Mills and Drance[12] compared the relationship between visual disability, as assessed by their questionnaire, and both visual acuity and visual field, as determined by the Esterman visual field score[15] in patients with severe glaucoma; they reported that visual disability correlated more strongly with visual field than with visual acuity. Parrish et al[20] and Gutierrez et al,[21] however, reported that quality of vision in glaucoma patients, assessed by existing questionnaires,[9,11] correlated with both visual acuity and remaining visual field as assessed by either Esterman visual field score[15] or advanced glaucoma intervention study score.[22] Because these studies used different subjects and methods to assess visual disability and visual field damage, it is difficult to compare the results directly. Our results, however, demonstrating a significant correlation between visual disability and both visual acuity and visual field in the lower hemifield within 5° of the fixation is compatible with the latter report.[20,21]

The Esterman visual field score was proposed as a mechanism to assess the functional visual field. The assessment design is based on the theory that the central part of the field is functionally more important than the periphery, with the lower hemifield being more used than the upper.[15] The lower central field is more important for quality of vision than any other regions evaluated in the Esterman visual field score. Our present result with Japanese glaucoma patients agrees with this theory, indicating that the lower central visual field is also crucial for quality of vision for those who write and read vertically. In addition, no previous report has compared the relationship between visual disability and the

mean retinal sensitivity within subdivided regions of the central visual field. This study demonstrated that the lower hemifield within 5° of fixation is important for quality of vision in glaucoma patients. It remains possible that the importance of this region in our study results from the selection of subjects, Japanese glaucoma patients who read and write vertically. If a similar investigation would be performed in a Western nation, other regions within the central 30° visual field may be more crucial for good quality vision.

In open-angle glaucoma, the lower central field often partially remains until the very late stages of the disease,[23] partially explaining why some glaucoma patients do not consult an ophthalmologist until late in the disease. Late in disease, the lower central field is more depressed in POAG eyes than in NTG eyes.[24] In addition, glaucomatous visual field damage tends to be more severe within the lower central field in myopic patients.[25] According to our results, a more aggressive treatment of Japanese POAG patients with myopia is needed in earlier stages of disease before damage encroaches on the lower central field.

Appendix

We have summarized the methods by which we revised the preliminary questionnaire and assessed the internal consistency and reliability of the revised questionnaire below.

Subjects

One hundred two patients with glaucoma (48 patients with POAG, 42 patients with NTG, and 12 patients with primary angle-closure glaucoma) were examined. The mean age of patients was 62.5 ± 12.7 years. The mean best-corrected visual acuity by the logarithm of the minimum angle of resolution was 0.2 ± 0.4 in the better eye and 0.8 ± 0.8 in the worse eye. The MD of HFA 30-2 was −11.9 ± 5.9 dB in the eye with better MD and −16.8 ± 6.5 dB in the eye with worse MD.

Methods

Visual disability in 102 patients was assessed using a preliminary questionnaire originally written in Japanese, consisting of a total of 35 questions addressing 7 items (Table 1).

Analysis of the principal components was used to revise the preliminary questionnaire to exclude factors other than visual disability.

The internal consistency of the revised questionnaire was evaluated by calculating the item-to-total correlation and Cronbach's α for each item addressed in the revised questionnaire.

Using the revised questionnaire, visual disability was assessed twice at intervals of 1 month; the test–retest reproducibility of the revised questionnaire was evaluated by using the κ coefficient. Reliability of the revised questionnaire was examined by calculating the mean and standard deviation of the differences in total DI from the first (total DI-1) and second (total DI-2) assessments. A power analysis was also performed.

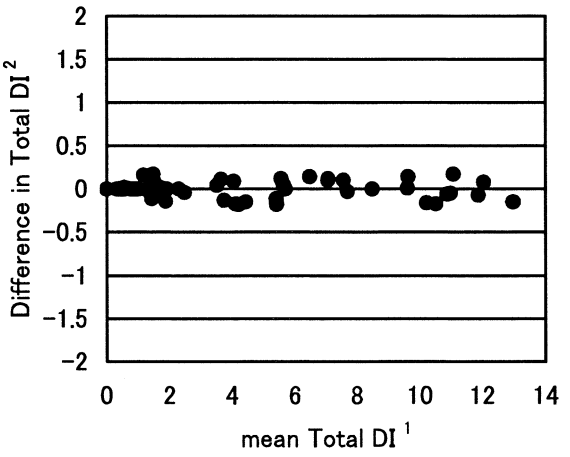


Figure 3. Difference against the mean for repeated total disability index (DI) measurements. [1]Mean total DI of the first (total DI-1) and second (total DI-2) assessment. [2]Differences in total DI between total DI-1 and total DI-2.

Results

The eigenvalues for principal components 1, 2, and 3 were 19.3, 2.4, and 1.6, with proportions of 0.6, 0.07, and 0.05, respectively. Because only principal component 1 demonstrated a high eigenvalue and a high proportion among these components, we suggest the existence of single principal component. All the questions included in the preliminary questionnaire exhibited high principal component weight for principal component 1 (0.54 to 0.85 for principal component 1, −0.30 to 0.57 for principal component 2, −0.34 to 0.38 for principal component 3). Five questions (written in italics in Table 1), however, also demonstrated a high principal component weight for principal component 2 (0.51 to 0.55), suggesting the possibility of including a factor other than visual disability. These five questions were excluded from the preliminary questionnaire.

In the revised questionnaire, the item-to-total correlation ranged from 0.81 to 0.91, whereas Cronbach's α varied between 0.85 and 0.88.

The κ coefficient was 1.0 (standard error, 0.09 to 0.13) for 16 questions and 0.9 (standard error, 0.10 to 0.13) for 14 questions. The mean score of total DI-1 and total DI-2 were 4.05 ± 3.84 and 4.06 ± 3.85, respectively. The difference between total DI-1 and total DI-2 was −0.01 ± 0.09 (range, −0.18 to 0.17) (Fig 3).[26] The sample size of 100 patients should allow the detection of a 1.54 difference in total DI-1 with $\alpha = 0.05$ and $\beta = 0.20$.[27]

Conclusion

The internal consistency and reliability of the revised questionnaire were found to be satisfactory.

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